



Dr Shubham Sahay with a few members of NeuroCHaSe Research Group on the IIT Kanpur campus

sation, you're able to process it and, at the same time, some part of it is going into your memory. You're able to recall something from the conversation. You encounter different types of smells in your surroundings, but you're able to identify which is coming from which place. That is because you have such an efficient processing engine—your brain. In neuromorphic computing, you basically take inspiration from the principles of the brain and try to mimic those on hardware utilising knowledge from nanoelectronics and VLSI.

In semiconductor electronics, the passage of information takes place with the help of electrons. Electrons are fundamental particles that travel very fast. But in your brain, it is the calcium ions that carry the information from one neuron to another neuron via synaptic clefts. For ions to flow you must create a concentration difference, which makes it a very slow process. But even with such a slow process, you can perform everything so efficiently in real time and with such a small

amount of energy. It is possible only because of the massive parallelism that our brain offers; computations are performed parallelly.

Even with its lower operating speed, neuromorphic computing is superior to what we're currently using in our systems, and that's only because of its inherent architecture.

SO, WE WERE
EXPLOITING
FLASH MEMORY
TO DEVELOP
VECTOR-BY-MATRIX
MULTIPLICATION
ACCELERATORS
WHICH CAN BE USED
TO PROCESS THE
DATA AT THE SAME
PLACE WHERE IT IS
STORED.

So how do we accomplish this?

We need to realise something like the processing elements in the brain on hardware, and memristors are one of the potential candidates (it would not be an exaggeration if I say most promising candidates) for realising processing entities such as neurons and synapses. Memristors basically can mimic synaptic elements. Apart from memristors, there are memtransistors as well, which are three-terminal devices, such as ferroelectric FETs, flash memories, etc. Nowadays, you also have all sorts of exotic materials coming into the picture—you have various devices showing neuron/synapse-like properties made up of organic substrates and even flexible substrates. But the most promising candidate, considering the present technical knowhow, is the memristor because it can be integrated with the present technology as well.

In theory, memristors existed in the 1960s itself, when a professor of UC Berkeley (Prof. Leon O. Chua) found out this missing element—